

Continuity



Testing continuity

- 1 Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.
 - a Find $\lim_{x \rightarrow 2^-} f(x)$.
 - b Find $\lim_{x \rightarrow 2^+} f(x)$.
 - c Is f continuous at $x = 2$? Justify.
 - 2 The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k .
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Types of discontinuity and the Intermediate Value Theorem

- 3 Classify the discontinuity of each function at the given point:
 - a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$
 - b $f(x) = \frac{1}{x - 2}$ at $x = 2$
 - c $f(x) = \lfloor x \rfloor$ at $x = 3$
 - 4 Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.
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Continuity on an interval

- 5 State the largest interval(s) on which each function is continuous:
 - a $f(x) = \frac{1}{x - 4}$
 - b $f(x) = \sqrt{x - 3}$
 - c $f(x) = \ln(x + 2)$
 - 6 Find the value of c that makes $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$, and $f(3) = c$, continuous at $x = 3$.
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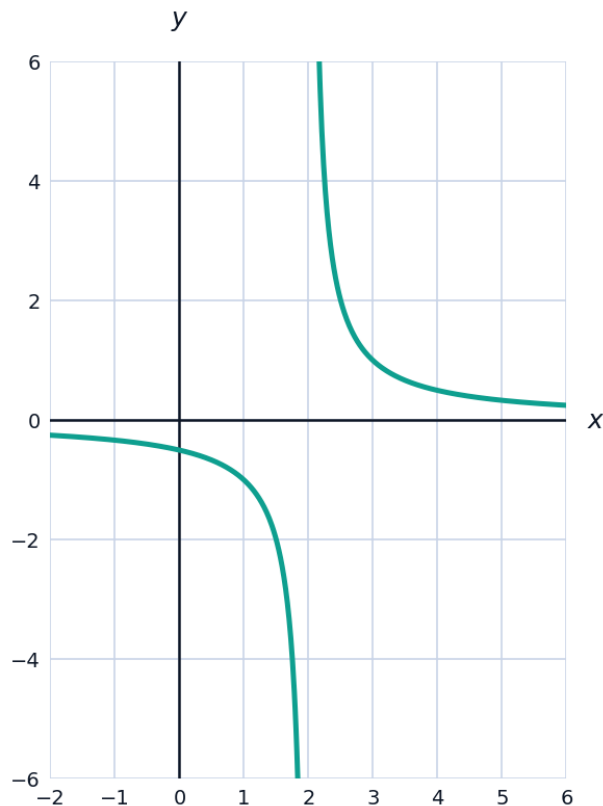
Applying the Intermediate Value Theorem

- 7 Show that $f(x) = \cos x - x$ has a root in the interval $(0, 1)$.

- 8 A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.
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Visualizing discontinuities

- 9 The function $f(x) = \frac{1}{x-2}$ (with an infinite discontinuity at $x = 2$) is graphed below. Use the graph to describe the one-sided behavior near $x = 2$.



Continuity of composite and combined functions

- 10 Explain why the composition of two continuous functions is continuous, and use this to argue that $h(x) = \sqrt{x^2 + 1}$ is continuous everywhere.
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Removable discontinuities and redefinition

- 11 The function $f(x) = \frac{x^2 + x - 6}{x - 2}$ has a removable discontinuity at $x = 2$. Find the value that, if assigned to $f(2)$, would make f continuous there.
- 12 Find all values of x where $f(x) = \frac{x^2 - 16}{x^2 - 2x - 8}$ has a discontinuity, and classify each as removable or non-removable.

The Extreme Value Theorem (a continuity consequence)

- 13** State the Extreme Value Theorem, and explain why $f(x) = x^2$ is guaranteed to have an absolute maximum and minimum on $[-3, 2]$.